

TEMPEST Model Bake-Off

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Abstract

This report will document a mathematical and computational closure program spanning particle-to-kinetic, wave-to-kinetic, kinetic-to-fluid, and application-scale reductions motivated by TEMPEST. It will compare classical Markov closures, explicit cumulant and memory models, physics-informed neural networks, neural operators, physics-informed neural operators, learned closures, structured learned dynamics, and action-conditioned world models. The experiment is organized as a linked sequence of effective-interaction, particle-to-kinetic, kinetic-to-moment, and application-transfer tests rather than a single architecture leaderboard. The application profiles are a dusty-plasma many-body closure ladder and a core-collapse-supernova turbulence closure that transfers unresolved 3D effects into a reduced 1D carrier.

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1 Introduction

The organizing question is not which architecture wins in the abstract, but what information is required to close a reduced model reliably and how that requirement changes across asymptotic and breakdown regimes.

We built an evaluation harness for computational closure methods: a fixed suite of test problems and forcing regimes, a common interface through which any closure model—analytic, learned, or hybrid—is invoked, and a standardized battery of accuracy, stability, and cost metrics computed uniformly across candidates. The harness separates the science (the closures) from the scaffolding (data handling, integration, scoring), so that adding a new method requires implementing one interface, and every reported comparison is reproducible and fair by construction.

2 Mathematical closure framework

This section will define scale transitions and exact hierarchy defects, connected correlations and cumulants, interaction histories and finite memory, and the decomposition of numerical, statistical, asymptotic, closure, and learning error.

For a representative hierarchy,

$$f_2 = f_1 f_1 + g_2, \quad \partial_t f_1 = \mathcal{L}f_1 + \mathcal{C}[f_1] + \mathcal{R}[g_2, g_3, \dots], \quad (1)$$

where the closure problem is to approximate the residual $\mathcal{R}$ from a declared information set while preserving the required mathematical structure.

3 Canonical scale-limit benchmarks

The canonical track is motivated by the long-time scale-limit program of Zaher Hani and collaborators. It contains hard-particle dynamics to Boltzmann, nonlinear wave dynamics to the wave kinetic equation, and Boltzmann dynamics to Euler and Navier–Stokes–Fourier. These are not ready-made TEMPEST closures. They are benchmark regimes in which the microscopic and reduced equations, asymptotic coordinates, expected limit, and discarded information are all explicit.

This makes them the validation layer for the computational evaluation harness. The particle track exposes connected cumulants, recollisions, and collision ancestry; the wave track exposes spectral cumulants and resonant interaction histories; and the kinetic-to-fluid track exposes non-Maxwellian moments and constitutive remainders. Each benchmark will test the trusted asymptotic regime, controlled breakdown outside its assumptions, long-time behavior, and separation of numerical, ensemble, asymptotic, and closure-model errors. TEMPEST applications retain this diagnostic protocol but replace the equations, states, constraints, observables, and coefficients.

4 Experimental pipeline

For every transition the experiment declares a high-fidelity state X , a projection Π , a retained state $u = \Pi X$, and a trusted reduced operator F_0 . The exact or estimable closure defect is

$$C^*[X] = \Pi F(X) - F_0(\Pi X). \quad (2)$$

The specification names the discarded particles, fields, correlations, moments, and histories whose effect appears in C^* , together with the information available to each candidate during training and inference.

The complete execution sequence is:

1. predeclare states, scaling coordinates, forcing regimes, splits, observables, information budgets, uncertainty estimators, and gates;

2. verify high-fidelity and reduced references under numerical refinement and mandatory physical checks;
3. generate versioned ensembles and project them into marginals, moments, cumulants, interaction histories, and closure defects;
4. fit matched analytic, learned, and hybrid candidates through the common model interface;
5. test the learned object a priori on held-out targets and then embed closures in the carrier solver for a-posteriori stability and feedback tests;
6. run predeclared asymptotic, controlled-breakdown, out-of-distribution, and cross-profile experiments without test-time retuning; and
7. audit every claim, retain failed runs, decompose numerical, sampling, estimator, asymptotic, model, and rollout error, and generate the report from machine-readable results.

4.1 Four linked bake-offs

The first stage learns an effective force at the retained-particle level after ions, electrons, wakes, fields, collisions, or faster dynamics have been eliminated. It compares analytic interaction laws, identifiable pair kernels, physics-tailored force decompositions, contextual graph models, and finite-memory auxiliary-state models. This is a closure when the force represents eliminated environmental degrees of freedom; it is not yet a kinetic or fluid closure.

The second stage projects verified particle ensembles to a one-particle distribution. It compares classical mean-field or kinetic operators, explicit correlation and cumulant corrections, weak-form discovered operators, learned kinetic residuals, memory models, and neural operators or PINOs when paired solution families are available. Particle cumulants, factorization error, hierarchy residuals, collective modes, transport, and long kinetic rollouts define the evidence.

The third stage projects kinetic solutions to density, momentum, energy, and a predeclared set of additional moments. Classical constitutive laws, Grad or maximum-entropy closures, hyperbolic learned moment closures, explicit cumulant and memory states, and structure-preserving learned residuals are tested on stress and heat flux, realizability, hyperbolicity, conservation, entropy behavior, classical-limit recovery, and embedded-solver stability.

The fourth stage transfers only validated diagnostic logic, state-selection rules, structural parameterizations, uncertainty procedures, and stopping criteria to a TEMPEST application. Coefficients, weights, state variables, and theorem assumptions remain application-specific unless a separate experiment justifies their transfer.

4.2 Cross-cutting pair, context, and memory experiment

The proposed dusty-plasma reference family contains heterogeneous particles, confinement, drag, screened repulsion, nonreciprocal wake effects, external forcing, and an optional dynamical wake or field coordinate ζ_j :

$$m_i \dot{v}_i = F_i^{\text{conf}} + F_i^{\text{ext}} - \gamma_i v_i + \sum_{j \neq i} F_{ij}(r_i, r_j, \zeta_j; \mu) + \eta_i, \quad \tau_w \dot{\zeta}_j = \zeta_{\text{eq}}(\mathcal{E}_j) - \zeta_j. \quad (3)$$

The actual reference physics must be selected with the scientific owner. The design sweeps wake relaxation τ_w , density or coupling, screening, heterogeneity, and forcing history. The same data

compare an analytic pair law, learned pair kernel, instantaneous many-particle graph model, finite-memory graph or auxiliary-state model, and unrestricted black-box control. The experiment locates where pairwise information stops being sufficient, where many-particle context becomes necessary, and where explicit memory earns its cost. Projected cumulants and interaction-history complexity are tested as early warnings of those boundaries. The resulting ensembles then feed the particle-to-kinetic and kinetic-to-moment stages.

5 TEMPEST application profiles

The dusty-plasma profile retains observed particles while eliminating plasma, wake, field, and sub-observation-time degrees of freedom. Its nested pair, contextual graph, and finite-memory candidates locate when each information class becomes necessary. The same verified ensembles then supply particle-to-kinetic and kinetic-to-moment closure targets.

The second science application is core-collapse-supernova turbulence. A 3D radiation hydrodynamics or (M)HD reference is projected by a declared angular or volume average, Favre/Reynolds convention, radial binning, filter, and shock alignment into a reduced 1D carrier. The exact closure defect is

$$C_{\text{SN}}^*[X] = \langle F_{\text{3D}}(X) \rangle_{\Omega} - F_{\text{1D}}(\langle X \rangle_{\Omega}), \quad (4)$$

decomposed into Reynolds stress or turbulent pressure, turbulent energy and enthalpy flux, production, dissipation, mixing, and optional finite-memory terms. The initial candidates are an unaugmented 1D carrier, a declared local RANS/mixing-length or STIR-like baseline, explicit turbulent-state augmentations, structure-preserving learned residuals, and finite-memory corrections. Each is tested after embedding on shock-radius evolution, revival/runaway timing, turbulent support, conservative energy and lepton accounting, realizability, explosion energy, and nucleosynthesis-sensitive thermodynamic histories.

6 Methods and information budgets

Methods will include classical, cumulant, and memory closures; PINNs, neural operators, and PINOs; learned closures and structured dynamics; and action-conditioned world models. Privileged information and matched comparisons will be reported explicitly. PINNs are evaluated as equation-informed approximations of particular forward or inverse solutions; neural operators and PINOs target parameterized solution maps; learned closures target an identified residual inside a retained solver; and a world model is admitted only for a genuine sequential intervention. Direct rankings are restricted to candidates approximating the same object.

7 Experimental protocol

The protocol will cover reference-solver verification, ensemble and estimator uncertainty, a-priori closure tests, a-posteriori embedded-solver tests, and transfer and failure-boundary experiments. Separate scorecards report target accuracy and identifiability, dynamics and reliability, closure diagnosis, physical and mathematical validity, limit recovery and transfer, and total information and computational cost. Unlike tasks are not averaged into one overall winner.

8 Results

Results will report recovery of classical limits; cumulant and memory diagnostics; predictive and embedded closure performance; dusty-plasma and supernova application-profile performance; and compute and information efficiency.

9 Discussion

The discussion will ask which closure information is necessary, what transfers from canonical and dusty-plasma systems to the supernova profile, and where negative results limit generalization.

10 Reproducibility statement

The reproducibility record will link data and result manifests, code, configurations, environments, claim levels, and retained failures.